

Chat Application @ Unified Mentor

Demo

<https://winter-f3cb5.web.app>

Overview

This project is a realtime chat application, built to serve as a reusable template for messaging-based applications. Emphasizing on firebase's full-stack capabilities, providing Backend-as-a-Service (BaaS) for realtime communication, authentication and data-persistence.

Tech Stack

- NextJs (Typescript) + Tailwind + Shadcn UI
- Firebase
 - Firebase Auth
 - Firestore
 - Firebase Hosting
 - Firebase Functions

Features

1. **Authentication:** Supports email+password and google authentication.
2. **Realtime Messaging:** Supports realtime messaging between users.
3. **Rooms:** Supports creating and joining rooms.
4. **Profile:** Supports updating profile information.

Technical Details

Multi-tenancy

Multi tenancy was chosen to allow future expansion into multi-organizational applications.

Database Schema (Firestore)

```
tenants (collection)
  -- tenant_id (document)
    -- users (sub-collection)
      -- user_id (document)
        -- uid (string)
        -- email (string)
        -- displayName (string)
        -- photoURL (string)
        -- phoneNumber (string)
        -- createdAt (timestamp)
        -- updatedAt (timestamp)
    -- rooms (sub-collection)
      -- room_id (document)
        -- id (string)
        -- name (string)
        -- description (string)
        -- ownerId (string)
        -- createdAt (timestamp)
        -- members (array of users)
    -- messages (sub-collection)
      -- chat_id / room_id (document, compound key)
        -- conversation (sub-collection)
          -- message_id (document)
            -- senderId (string)
            -- content (string)
            -- senderName (string)
            -- timestamp (timestamp)
            -- type (string)
```

UI/UX Highlights

- Responsive layout with full viewport height
- Content area adjusts to available space without overflow
- Rounded edges for a softer look
- Dynamic theme switching using `next-themes`
- Consistent color palette using Shadcn UI and TailwindCss